

Intracranial hypertension and herniation

CLINICAL SIGNS OF HERNIATION

- **Motor decline:** Spontaneous GCS motor score decrease of ≥ 1 point.
- **Pupillary reactivity:** Decrease in pupillary reactivity (Neurological Pupil Index, NPi < 3).
- **Asymmetry:** New pupillary asymmetry or unilateral dilation (ipsilateral mydriasis).
- **Focal deficit:** New focal motor deficit or abnormal posturing (decorticate / decerebrate).
- **Cushing's Triad (Late Sign):** Systolic hypertension, bradycardia, and irregular respirations. ***Cushing triad is a LATE sign of brainstem compression.***

Syndrome	Anatomical Substrate	Clinical Exam & Diagnostic Trap
Uncal (Lateral)	Medial temporal lobe (uncus) pushed over tentorial edge	Ipsilateral sluggish/dilated pupil (CN III compressed), contralateral hemiparesis. <i>Trap: Kernohan's Notch</i> causes false-localizing ipsilateral hemiparesis.
Central (Axial)	Downward diencephalic and midbrain displacement	Progressive stupor, midpoint fixed pupils, decorticate to decerebrate posturing. <i>Trap: Symmetrical signs</i> often confused with metabolic encephalopathy.
Subfalcine (Cingulate)	Cingulate gyrus displaced under the falx cerebri	Often clinically silent, or presents with contralateral lower extremity weakness. <i>Trap: ACA compression</i> causes frontal/leg territory infarction.
Tonsillar (Downward)	Cerebellar tonsils forced through the foramen magnum	Cushing's triad, flaccid quadriplegia, respiratory arrest. <i>Trap: Cushing triad</i> is a LATE sign of brainstem compression.

MANAGEMENT (TIERED PATHWAY)

Tier 0: Fundamental Optimization (For all at-risk patients)

- Elevate HOB 30°; strict neutral midline neck alignment to preserve venous outflow.
- Euvolemia (isotonic saline; avoid hypotonic \$D_5W\$), normothermia (treat fevers), normocapnia (\$pCO_2\$ 35–40 mmHg).

Tier 1: Initial Interventions

- Continuous CSF drainage via EVD (typically set at 10–15 cmH₂O).
- Analgesia/sedation (propofol/fentanyl) to prevent coughing, agitation, or ventilator dyssynchrony.

Tier 2: Hyperosmolar & Ventilatory Escalation

- **Mannitol 20%:** 1 g/kg IV bolus over 20–30 min. Must use in-line filter. **Hold if Serum Osmolarity > 320 mOsm/kg or Gap ≥ 20 mOsm/kg.**
- **Hypertonic Saline (HTS):** 3% (150–250 mL bolus) or 23.4% (30 mL rescue bolus; central line access only). **Hold if Serum Na > 155 mEq/L or Cl > 115 mEq/L.**
- **Controlled Hyperventilation:** Target \$pCO_2\$ 30–35 mmHg. Temporary bridge only; avoid prolonged use or \$pCO_2 < 30\$ (ischemia risk).

Tier 3: Refractory & Salvage Measures

- High-dose barbiturate therapy (pentobarbital) titrated to burst suppression on EEG.
- **Decompressive Surgery:**
 - **Malignant MCA (DHC):** Age ≤ 60 years, GCS decline ≥ 1 / pupil changes, infarct $\geq 50\%$ MCA, within 48h (DECIMAL/DESTINY).
 - **Cerebellar Stroke:** Suboccipital craniectomy for brainstem compression, 4th ventricle effacement, or hydrocephalus.

TIER 0: BASE

hob 30°, midline, isotonic

TIER 1: INITIAL

evd drainage, sedation

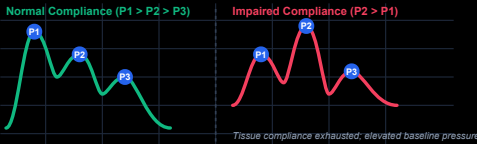
TIER 2: MEDICAL

osmotherapy, bridge vent

TIER 3: SALVAGE

barb coma, craniectomy

ICP WAVEFORM ANALYSIS



- **P1 (Percussion wave):** Arterial pulsation.
- **P2 (Tidal wave):** State of intracranial compliance (elastic reserve).
- **P3 (Dicrotic wave):** Venous pulsations.
- **Compliance States:**
 - **Normal Compliance:** $P1 > P2 > P3$ (elastic brain tissue easily cushions pulsations).
 - **Impaired Compliance / High ICP:** $P2 > P1$ (brain tissue reserve exhausted; high risk of herniation).

CPP = MAP - ICP

In patients with intracranial hypertension or mass effect, cerebral perfusion is highly pressure dependent, and cautious BP lowering is recommended.